

Ex. No. 2	Working with HTML5 Form and Media Elements
Date of Exercise	24 July 2026
GitHub Host Link	https://jinto-joseph.github.io/WEB_TECH_LAB/24_July_Exp_2/exercise1.html

Aim

To design and develop a responsive webpage using HTML5 form controls, multimedia elements (<audio>, <video>), structural grouping elements (<fieldset>, <legend>), progress bars, and embedded frames (<iframe>).

Description

HTML Form Elements

All forms start with the <FORM> tag and end with </FORM>. All other form objects go between these two tags.

The form tag has two main properties: METHOD and ACTION.

```
<form action="/action_page.php" method="get"> </form>
```

The `<form>` element can contain one or more of the following form elements:

- <input>
- <textarea>
- <button>
- <select>
- <option>
- <fieldset>
- <label>

The `<input>` tag specifies an input field where the user can enter data. `<input>` elements are used within a `<form>` element to declare input controls that allow users to input data. An input field can vary in many ways, depending on the `type` attribute.

```
<form action="/action_page.php" method="get">
```

```
  First name: <input type="text" name="fname"><br>
```

```
  Last name: <input type="text" name="lname"><br>
```

```
<input type="submit" value="Submit">
```

</form>

Input Types:

Here are the different input types you can use in HTML

```
<input type="button">
<input type="checkbox">
<input type="date">
<input type="datetime-local">
<input type="email">
<input type="file">
<input type="hidden">
<input type="image">
<input type="month">
<input type="number">
<input type="password">
<input type="radio">
<input type="range">
<input type="reset">
<input type="search">
<input type="submit">
<input type="tel">
<input type="text">
<input type="time">
<input type="url">
<input type="week">
```

Textarea Element

The ``<textarea>`` tag defines a multi-line text input control. A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier). The size of a text area can be specified by the ``cols`` and ``rows`` attributes, or even better, through CSS' ``height`` and ``width`` properties.

```
<textarea rows="4" cols="50">
</textarea>
```

Select Element

The ``<select>`` element is used to create a drop-down list. The ``<option>`` tags inside the ``<select>`` element define the available options in the list.

```
<select>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="mercedes">Mercedes</option>
  <option value="audi">Audi</option>
</select>
```

Fieldset Element

The ``<fieldset>`` tag is used to group related elements in a form. The ``<fieldset>`` tag draws a box around the related elements.

```
<form>
  <fieldset>
    <legend>Personalia:</legend>
    Name: <input type="text"><br>
    Email: <input type="text"><br>
    Date of birth: <input type="text">
  </fieldset>
</form>
```

Media Elements:

The following are the most popular media elements.

- **audio:**

Audio element is used for defining or creating a music element in your simple HTML page.

```
<audio controls>
  <source src="1.mp3" type="audio/mpeg">
</audio>
```

- **video:**

The Video element creates a video element in your HTML page.

```
<video controls>
  <source src="1.mp4" type="video/mp4">
</video>
```

- **embed:**

It is also called a container because as the name suggests it is used for defining containers for the external applications or we can say plug-ins for the applications.

```
<embed src="1.swf">
```

Program:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Ex.No: 02 - Working with HTML5 Form and Media Elements</title>
  <style>
    :root {
      --primary-color: #2b5797;
      --border-color: #cccccc;
      --bg-light: #f9f9f9;
      --text-color: #333333;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: Arial, Helvetica, sans-serif;
      background-color: #f4f6f8;
      color: var(--text-color);
      padding: 20px;
      display: flex;
      justify-content: center;
    }

    .wrapper {
      width: 100%;
      max-width: 960px;
      background-color: #ffffff;
      border: 1px solid #b5b5b5;
      padding: 20px;
      box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
      display: flex;
      flex-direction: column;
      gap: 20px;
    }

    .header {
      display: flex;
```

```
        justify-content: space-between;
        align-items: center;
        padding-bottom: 15px;
        border-bottom: 1px solid var(--border-color);
    }

    .profile-img {
        width: 120px;
        height: 140px;
        object-fit: cover;
        border: 1px solid var(--border-color);
        border-radius: 4px;
    }

    .header h1 {
        font-size: 26px;
        color: #111111;
        font-weight: bold;
    }

    .navbar {
        display: flex;
        justify-content: space-between;
        align-items: center;
        padding: 10px 0;
        border-bottom: 1px solid var(--border-color);
    }

    .nav-links {
        display: flex;
        gap: 40px;
    }

    .nav-links a {
        color: #551a8b;
        text-decoration: underline;
        font-weight: bold;
        font-size: 15px;
    }

    .search-box input[type="search"] {
        padding: 4px 8px;
        border: 1px solid #aaa;
        border-radius: 2px;
    }

    .grid-top {
        display: grid;
        grid-template-columns: 240px 1fr;
        gap: 20px;
```

```
        align-items: start;
    }

    .skills-list {
        display: flex;
        flex-direction: column;
        gap: 14px;
    }

    .skill-item label {
        display: block;
        font-weight: bold;
        font-size: 13px;
        margin-bottom: 4px;
    }

    progress {
        width: 100%;
        height: 18px;
        appearance: none;
        -webkit-appearance: none;
        -moz-appearance: none;
        border: 1px solid #777777;
        border-radius: 3px;
        background-color: #e0e0e0;
        overflow: hidden;
        display: block;
    }

    progress::-webkit-progress-bar {
        background-color: #e0e0e0;
        border-radius: 3px;
    }

    progress::-webkit-progress-value {
        background: linear-gradient(180deg, #62c462 0%, #468847 100%);
        border-radius: 2px;
    }

    progress::-moz-progress-bar {
        background: linear-gradient(180deg, #62c462 0%, #468847 100%);
        border-radius: 2px;
    }

    .media-container {
        display: flex;
        flex-direction: column;
        gap: 10px;
    }
```

```
video {
  width: 100%;
  height: 220px;
  border: 1px solid #ccc;
  background-color: #000000;
  border-radius: 4px;
  object-fit: contain;
}

audio {
  width: 100%;
  margin-top: 5px;
}

.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
}

fieldset {
  border: 1px solid #999999;
  padding: 15px;
  border-radius: 2px;
  background-color: #ffffff;
}

legend {
  font-weight: bold;
  padding: 0 6px;
  color: #222222;
  font-size: 14px;
}

.form-group {
  display: flex;
  align-items: center;
  margin-bottom: 12px;
  font-size: 13px;
}

.form-group label {
  width: 120px;
  font-weight: normal;
  color: #222222;
}

.form-group input[type="text"],
.form-group input[type="password"],
.form-group input[type="number"],
```

```
.form-group input[type="tel"],
.form-group input[type="email"],
.form-group input[type="url"],
.form-group input[type="date"],
.form-group select,
.form-group textarea {
    flex: 1;
    padding: 4px 6px;
    border: 1px solid #767676;
    border-radius: 2px;
    font-size: 13px;
}

.required-star {
    color: black;
    margin-left: 4px;
    font-weight: bold;
}

.radio-group {
    display: flex;
    gap: 15px;
    align-items: center;
}

.submit-btn {
    background-color: #e1e1e1;
    border: 1px solid #767676;
    padding: 4px 16px;
    cursor: pointer;
    border-radius: 2px;
    font-size: 13px;
}

.submit-btn:hover {
    background-color: #d0d0d0;
}

.bottom-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 20px;
}

.embed-box, .iframe-box {
    border: 1px solid #ccc;
    height: 320px;
    overflow: hidden;
    border-radius: 4px;
    background-color: #ffffff;
}
```



```

    }

    iframe, embed {
        width: 100%;
        height: 100%;
        border: none;
    }

    .footer {
        border-top: 1px solid var(--border-color);
        padding-top: 15px;
        font-size: 13px;
        color: #333333;
    }

    .footer p {
        margin-bottom: 4px;
    }
</style>
</head>
<body>

    <!-- HTML Comment Tag Requirement -->
    <div class="wrapper">

        <header class="header">
            
            <h1>Mr.Jinto Joseph</h1>
        </header>

        <nav class="navbar">
            <div class="nav-links">
                <a href="#academics">Academics</a>
                <a href="#achievements">Achievements</a>
                <a href="#friends">Friends</a>
            </div>
            <div class="search-box">
                <input type="search" placeholder="search...">
            </div>
        </nav>

        <section class="grid-top">
            <div class="skills-list">
                <div class="skill-item">
                    <label for="photoshop">Photoshop:</label>
                    <progress id="photoshop" value="70"
max="100">70%</progress>

```

```

    </div>
    <div class="skill-item">
        <label for="java">Java:</label>
        <progress id="java" value="85" max="100">85%</progress>
    </div>
    <div class="skill-item">
        <label for="cpp">C++:</label>
        <progress id="cpp" value="90" max="100">90%</progress>
    </div>
    <div class="skill-item">
        <label for="php">PHP:</label>
        <progress id="php" value="60" max="100">60%</progress>
    </div>
    <div class="skill-item">
        <label for="javascript">Javascript:</label>
        <progress id="javascript" value="80"
max="100">80%</progress>
    </div>
</div>

<div class="media-container">
    <video controls style="width: 100%; height: 220px; border:
1px solid #ccc; border-radius: 4px; background: #000; object-fit:
contain;">
        <source src="video.mp4" type="video/mp4">
        <source
src="https://www.w3schools.com/html/mov_bbb.mp4" type="video/mp4">
        Your browser does not support the video tag.
    </video>

    <audio controls style="width: 100%; margin-top: 4px;">
        <source
src="https://www.soundhelix.com/examples/mp3/SoundHelix-Song-1.mp3"
type="audio/mpeg">
        Your browser does not support the audio element.
    </audio>
</div>
</section>

<form action="#" method="get">
    <div class="form-grid">

        <fieldset>
            <legend>Personal Information</legend>

            <div class="form-group">
                <label for="username">User name:</label>
                <input type="text" id="username" name="username"
placeholder="Enter the user name">
                <span class="required-star">*</span>

```

```
</div>

<div class="form-group">
  <label for="password">Password:</label>
  <input type="password" id="password"
name="password" value="Madangs-10">
  <span class="required-star">*</span>
</div>

<div class="form-group">
  <label for="regno">Reg No:</label>
  <input type="text" id="regno" name="regno"
value="URK24CS1210">
  <span class="required-star">*</span>
</div>

<div class="form-group">
  <label for="age">Age:</label>
  <select id="age" name="age">
    <option value="19-24" selected>19-24</option>
    <option value="18">18</option>
    <option value="25+">25+</option>
  </select>
</div>

<div class="form-group">
  <label>Gender:</label>
  <div class="radio-group">
    <label style="width:auto;"><input type="radio"
name="gender" value="Male"> Male</label>
    <label style="width:auto;"><input type="radio"
name="gender" value="Female" checked> Female</label>
  </div>
</div>

<div class="form-group">
  <label for="dob">DOB:</label>
  <input type="date" id="dob" name="dob"
value="2005-07-17">
  <span class="required-star">*</span>
</div>

<div class="form-group">
  <label for="favcolor">Favorite Color:</label>
  <input type="color" id="favcolor" name="favcolor"
value="#e6f2ff">
</div>

<div class="form-group">
```

```

        <label for="profile_photo">Change Profile
Photo:</label>
        <input type="file" id="profile_photo"
name="profile_photo">
    </div>
</fieldset>

<fieldset>
    <legend>Contact Information</legend>

    <div class="form-group" style="align-items:
flex-start;">
        <label for="address" style="margin-top:
4px;">Address:</label>
        <textarea id="address" name="address" rows="3"
placeholder="Your address"></textarea>
    </div>

    <div class="form-group">
        <label for="state">State:</label>
        <select id="state" name="state">
            <option value="">Select</option>
            <option value="Tamil Nadu">Tamil Nadu</option>
            <option value="Kerala">Kerala</option>
            <option value="Karnataka">Karnataka</option>
        </select>
        <span class="required-star">*</span>
    </div>

    <div class="form-group">
        <label for="zipcode">Zipcode:</label>
        <input type="number" id="zipcode" name="zipcode"
placeholder="only 6 numbers">
        <span class="required-star">*</span>
    </div>

    <div class="form-group">
        <label for="mobile">Mobile:</label>
        <input type="tel" id="mobile" name="mobile"
placeholder="( +91) 99999-99999">
    </div>

    <div class="form-group">
        <label for="email">Email:</label>
        <input type="email" id="email" name="email"
placeholder="valid email">
        <span class="required-star">*</span>
    </div>

    <div class="form-group">

```

```
        <label for="url">URL:</label>
        <input type="url" id="url" name="url"
placeholder="valid url">
        <span class="required-star">*</span>
    </div>

    <div class="form-group">
        <input type="checkbox" id="iverified"
name="iverified" checked style="margin-right: 8px;">
        <label for="iverified" style="width: auto;">I
verified</label>
    </div>

    <div class="form-group" style="margin-bottom: 0;">
        <input type="submit" value="Submit"
class="submit-btn">
    </div>
</fieldset>

</div>
</form>

<section class="bottom-grid">
    <div class="embed-box">
        <embed src="Ex2.pdf" type="application/pdf" width="100%"
height="100%">
    </div>

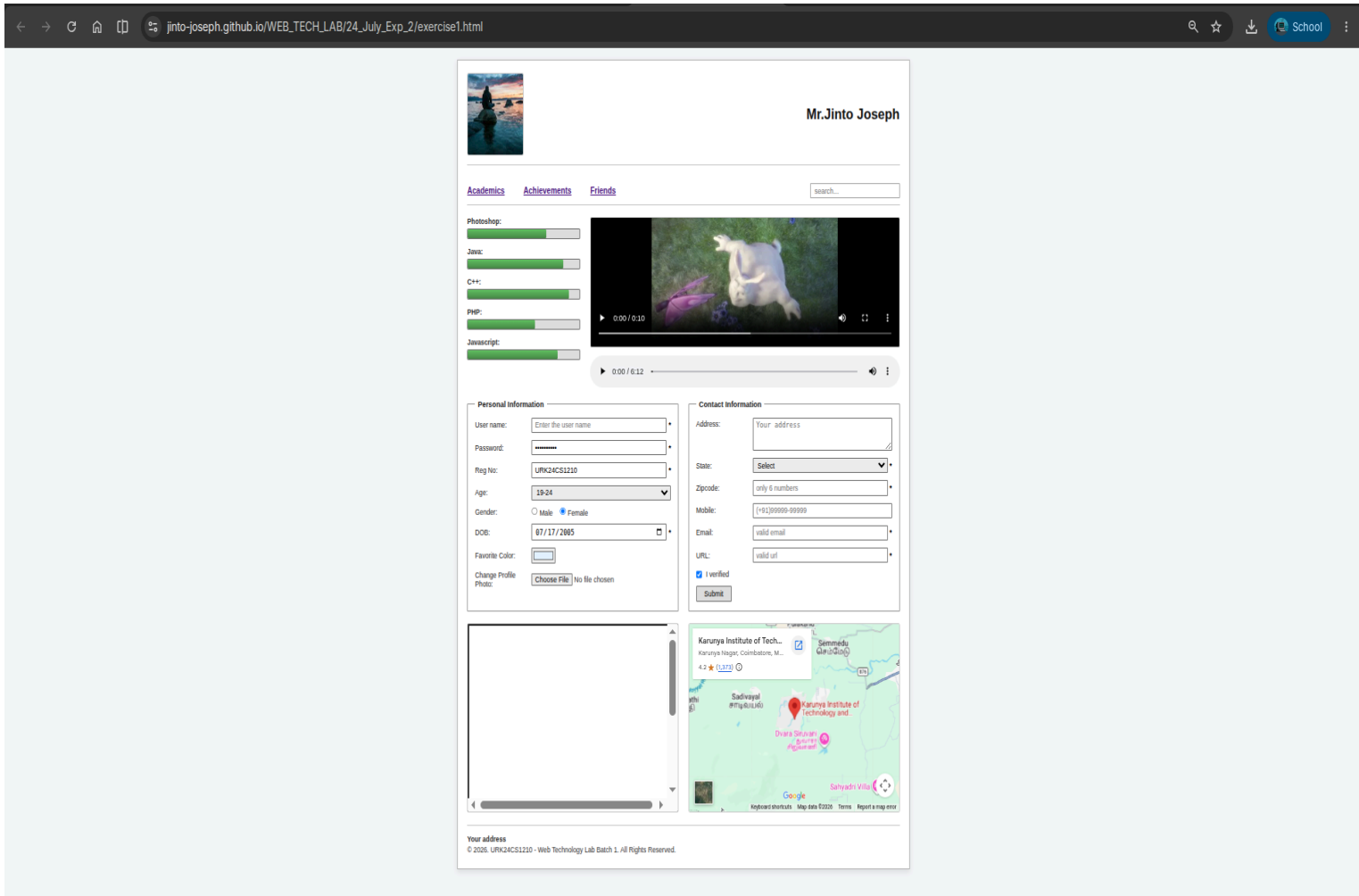
    <div class="iframe-box">
        <iframe
src="https://maps.google.com/maps?q=Karunya%20University,%20Coimbatore,%20
Tamil%20Nadu&t=&z=13&ie=UTF8&iwloc=&output=embed" loading="lazy"
title="Karunya University Location Map"></iframe>
    </div>
</section>

<footer class="footer">
    <p><strong>Your address</strong></p>
    <p>© 2026. URK24CS1210 - Web Technology Lab Batch 1. All Rights
Reserved.</p>
</footer>

</div>

</body>
</html>
```

Output:



Result:

The HTML5 webpage featuring form controls, media elements (<audio>, <video>), embedded frames (<iframe>), and skill progress bars was successfully designed, implemented, and verified according to the lab requirements.